

Paper	Dataset Used	Dataset Size	Model / Method	Evaluation Metrics	Key Results / Accuracy	Limitations / Gaps
Deep-learning-driven Cross-modal Image Fusion For Cartoon Captioning (Feng et al., 2025)	CartoonCap-9k, ToonFables-32k	CartoonCap-9k: 9,000 images with 5 captions each; ToonFables-32k: 32,000 images with 5 captions each	IBA-CD (Information Bottleneck-driven Cross-modal Alignment + Cascaded Diffusion Network)	BLEU-1, BLEU-4, METEOR, ROUGE, CIDEr	CartoonCap-9k: BLEU-1: 80.20, BLEU-4: 39.20, METEOR: 28.84, ROUGE: 56.32, CIDEr: 125.7; ToonFables-32k: BLEU-1: 67.74, BLEU-4: 29.31, METEOR: 22.73, ROUGE: 53.52, CIDEr: 65.40 BLIP-2 (fine-tuned): ROUGE-1: 0.508, ROUGE-2: 0.211, ROUGE-L: 0.373, Word Coverage: 0.464; Best humor combination: BLIP-2 + GPT-4	Computational complexity not discussed; Limited to two cartoon datasets; No humor evaluation (only literal captioning)
Humor Through Vision and Language (Singhal, 2024)	NYC Cartoon Dataset (New Yorker)	500 images used for fine-tuning; dataset contains image descriptions and captions	ViT (nlpcnnect/vit-gpt2), BLIP-2 (Salesforce/blip-large), LLMs: GPT-2, LLaMA, GPT-4, Mistral	ROUGE-1, ROUGE-2, ROUGE-L, BLEU Score, Word Coverage		Small fine-tuning dataset (500 images); No quantitative humor metrics; Manual evaluation only; Cultural bias in humor assessment
Inside Jokes: Identifying Humorous Cartoon Captions (Shahaf et al., 2015)	New Yorker Caption Contest (16 cartoons + 10 additional)	16 cartoons with 3,562-6,557 captions each; 10,160 pairwise rankings	Random Forest Classifier with linguistic features (perplexity, POS, readability, joke location, word embeddings)	Pairwise classification accuracy	Same-joke classifier: 69% accuracy; Different-joke classifier: 64% accuracy; Tournament ranking: 37% of shortlisted captions in top 10%	Small number of cartoons (16); Crowdfunder humor vs. New Yorker audience mismatch; Word embeddings miss capitalization-based jokes; No image features used
Learning Humor Through AI (Li, 2020)	New Yorker Caption Contest (NextML dataset)	182 contests, 1 million+ captions; sampled 10,739 captions for experiments	BERT (text only), BERT + VGG19 (multimodal), Random Forest Classifier	Classification accuracy, precision, recall, MSE (regression)	Classification: Failed — all captions predicted as unfunny due to class imbalance; Regression: MSE ~0.019 (minor improvement over baseline)	Extreme class imbalance (99.5% unfunny); Only 6 funny captions in sampled dataset; VGG19 pretrained on ImageNet (not cartoons); No successful classification achieved
Humor in AI: Massive Scale Crowd-Sourced Preferences (Zhang et al., 2024)	New Yorker Caption Contest	365 contests, 2.2 million captions, 250+ million human ratings	Models: Mistral-7B, LLaVA-7B, GPT-3.5 Turbo, GPT-4o, GPT-4o Vision, Claude-3-Opus; Alignment: SFT, RLHF (PPO), DPO, BoN	Overall Win Rate (%), Best Pick Win Rate (%), Average EAD (diversity), SBERT (diversity)	Best Overall Win Rate: Claude-3-Opus vs. Top 10: 54.40%, vs. Median: 88.46%; Best Pick Win Rate: GPT-4o Vision vs. Top 10: 47.80%; DPO Best Pick: 30.22% vs. 0-shot 12.64%; Human expert vs. Claude: Expert prefers Claude only 1.6%	Multimodal models underperform language-only models; RLHF shows limited improvement for humor; Humor is subjective; Reward models may not capture humor complexity